Jiong Xiao

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Education

MA - Cognitive Science September 2021 - May 2025
Distinction
University of Edinburgh
Main Courses: Computational Cognitive Science, Computational Cognitive Neuroscience, The Computational Mind, Seminar in Cognitive Modelling, Causal Cognition

MSc - Cognitive and Decision Sciences September 2025 - September 2026(expected)
University College London
Main Courses: Brain in Action, Advanced Statistics: Data Analysis and Modelling, Principles of Cognition, Neural Computation: Models of Brain Functions, Applied Deep Learning

Work Experience

Research Assistant
Oct 2022 - Jan 2023
Edinburgh, Scotland, United Kingdom
- Worked with Dr Barbora Skarabela on a project regarding polysemy and how children grasp a range of ambiguous meanings.
- Annotated word meanings in existing child language corpora.

Research Assistant
Oct 2023 - Apr 2024
Edinburgh, Scotland, United Kingdom
- Worked with Professor Hannah Rohde on an eye-tracking study investigating reading anticipation behaviour.
- Collected eye-tracking data for the study and coordinated the sessions with the participants.

Projects

Adaptive Control in Multidimensional Environments (Ongoing)

I worked on a project independently under the supervision of Dr Neil R Bramley. The project relates to control and function learning. Specifically, we investigated whether people tend to minimise the number of variables being controlled and the amount of changes made to the variable. I designed a novel task paradigm that investigates how people manipulate variables in complex environments. The task paradigm creates a large action space, allowing us to investigate how people use simplifying heuristics to facilitate action selection. The project sheds light upon how people select actions from a massive action space in light of resource-rationality through behavioural experiments along with computational modelling.

How Complexity Guides Mental Representations and Planning (Ongoing)

I am currently working on my master's project under the supervision of Professor Stephen Fleming, Dr Jason da Silva Castanheira and Dr Kevin O'Neil at the MetaLab led by Professor Stephen Fleming at UCL. My master's project focuses on how people simplify their representations of task environments. Specifically, we aim to conduct behavioural experiments examining how planning interacts with the construal of the environment. We are interested in how various kinds of complexity associated with the environment shape how humans build an internal model of the world. This sheds light upon the algorithmic basis of "data compression" in humans, as well as the connection between consciousness and cognition.

Honour and Awards

Simon Gray Prize 2022

The prize was awarded by the University of Edinburgh for outstanding performance in Logic 1.

Matthew Green Memorial Prize 2023

The prize was awarded by the University of Edinburgh to the student with the highest final marks in second-year psychology final tests.

Robert J. Glushko Prize 2025

The prize was awarded annually by the University of Edinburgh to one final-year student on the degree of MA Cognitive Science.

Publications

Xiao, J. and N. R. Bramley (2026). The benefits of staying local: Bounded adaptive control by modeling linearly and acting locally. In: *The 48th Annual Meeting of the Cognitive Science Society.* Austin, TX.

Skills

Programming	R, Python, Java, Haskell, HTML, Javascript
Languages	Mandarin(Native), English(Proficient)